

Exploratory Data Analysis Using SQL

Introduction

This project focuses on analyzing customer, order, product, and shop data using SQL. The purpose of the analysis is to understand sales performance, customer purchasing behavior, product demand, shop performance, revenue distribution, and other important business patterns.

The database used in this project contains four main tables: customers, orders, products, and shops. Different SQL queries were used to answer business-related questions and generate meaningful insights from the available data.

Objectives

The main objectives of this analysis are:

- To analyze customer purchasing behavior and spending.
 - To identify high-performing customers, products, and shops.
 - To study revenue and sales patterns.
 - To identify products that are not being ordered.
 - To understand customer loyalty and ordering behavior.
 - To compare shop performance across different locations and categories.
 - To generate a detailed customer summary for business analysis.
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Exploratory Data Analysis

Q1. Top 3 Highest Spending Customers

```
SELECT o.customer_id, CONCAT(c.first_name, ' ', c.last_name) AS customer_name,  
       round(SUM(o.total_amount),2) AS total_spent FROM sqlp_orders AS o  
JOIN sqlp_customers AS c  
ON c.customer_id = o.customer_id  
GROUP BY o.customer_id , c.first_name , c.last_name  
ORDER BY total_spent DESC  
LIMIT 3;
```

Purpose:

This analysis identifies the customers who have spent the highest amount on their orders. It helps the business recognize its most valuable customers.

Q2. Best-Selling Product in Each Category

```
SELECT *FROM
(
  (SELECT p.category,
    p.product_name,SUM(o.quantity) AS sold_quantity,
    DENSE_RANK() OVER (PARTITION BY p.category
      ORDER BY SUM(o.quantity) DESC) AS rnk FROM sqlp_products AS p
    JOIN sqlp_orders AS o
      ON o.product_id = p.product_id
    GROUP BY p.category, p.product_name) AS t
  WHERE rnk = 1;
```

Purpose:

This analysis identifies the most popular product within every product category. It helps understand which products perform best in different categories.

Q3. Customers Who Never Placed an Order

```
select c.customer_id, concat(c.first_name," ",c.last_name) as Full_name
from sqlp_customers as c
left join sqlp_orders as o
on c.customer_id = o.customer_id
where o.order_id is null;
```

Purpose:

This analysis identifies registered customers who have not placed any orders. This can help the business target inactive customers with offers or promotions.

Q4. Month-over-Month Revenue Growth for 2024

```

with monthly_revenue as ( select month(order_date) as month_, round(sum(total_amount)) as revenue
from sqlp_orders
where year(order_date) = "2024"
group by month(order_date))

select month_, revenue, lag(revenue) over(order by month_) as previous_amount,
round((revenue - lag(revenue) over(order by month_))/lag(revenue) over(order by month_)*100,2)
as persenateg
from monthly_revenue
order by month_;

```

Purpose:

This analysis is intended to measure how revenue changes from one month to another during 2024. It can help identify periods of business growth or decline.

Q5. Percentage of Cancelled or Returned Orders by Shop

```

WITH cr AS (
    SELECT
        shop_id,
        COUNT(order_id) AS cancel_orders
    FROM sqlp_orders
    WHERE order_status IN ('Cancelled', 'Returned')
    GROUP BY shop_id
)

SELECT
    s.shop_id,
    s.shop_name,
    (c.cancel_orders / COUNT(o.order_id)) * 100 AS percentage
FROM sqlp_shops AS s
LEFT JOIN cr AS c
    ON s.shop_id = c.shop_id
JOIN sqlp_orders AS o
    ON s.shop_id = o.shop_id
GROUP BY s.shop_id, s.shop_name, c.cancel_orders;

```

Purpose:

This analysis measures the percentage of cancelled and returned orders for each shop. It helps identify shops that may have higher order-related problems.

Q6. Customers Ordering from Multiple Shops

```
select c.customer_id,concat(c.first_name," ",c.last_name) as full_name,
count(distinct o.shop_id) as shop_id from sqlp_orders as o
join sqlp_customers as c
on o.customer_id = c.customer_id
group by c.customer_id,c.first_name,c.last_name
having shop_id >= 3
order by c.customer_id asc;
```

Purpose:

This analysis identifies customers who have ordered from multiple shops. It helps understand customers with broader purchasing activity across the available shops.

Q7. Second Highest Order Amount for Each Customer

```
select * from
(select customer_id,total_amount, dense_rank()
over(partition by customer_id order by total_amount desc)
as rnk
from sqlp_orders) t
where rnk = 2;
```

Purpose:

This analysis finds the second highest order value placed by each customer. It helps compare customers' higher-value purchases while excluding customers with only one order.

Q8. Running Total Revenue per Shop

```
SELECT shop_id, order_date, round(SUM(total_amount)) AS daily_revenue,  
round(SUM(SUM(total_amount))  
OVER (PARTITION BY shop_id ORDER BY order_date)) AS cumulative_revenue  
FROM sqlp_orders  
GROUP BY shop_id, order_date  
ORDER BY shop_id, order_date;
```

Purpose:

This analysis is intended to track how total revenue accumulates over time for each shop. It can help observe the revenue growth pattern of individual shops.

Q9. Products That Have Never Been Ordered

```
select p.product_id, p.product_name from sqlp_products as p  
left join sqlp_orders as o  
on p.product_id = o.product_id  
where o.product_id is null;
```

Purpose:

This analysis identifies products that have not received any orders. This can help the business review products with low or no customer demand.

Q10. Average Order Value by Loyalty Tier

```
select c.loyalty_tier, round(avg(o.total_amount),2) as Total_amount from sqlp_customers as c  
join sqlp_orders as o  
on c.customer_id = o.customer_id  
where c.signup_date < '2023-01-01'  
group by c.loyalty_tier;
```

Purpose:

This analysis compares the average order value across different customer loyalty tiers. Only customers who signed up before the specified date are included in the analysis.

Q11. Time Taken by Customers to Place Their First Order

```
select c.customer_id, concat(c.first_name," ",c.last_name), c.signup_date,
min(o.order_date) as first_order_date,
datediff(min(o.order_date), c.signup_date) as day_to_first_order
from sqlp_customers as c
join sqlp_orders as o
on c.customer_id = o.customer_id
GROUP BY c.customer_id,c.first_name,c.last_name,c.signup_date;
```

Purpose:

This analysis measures the number of days between customer signup and their first order. It helps understand how quickly customers start purchasing after registration.

Q12. Classification of Orders by Value

```
select order_id,total_amount,
case
when total_amount < 5000 then "Low"
when total_amount <= 20000 then "Median"
else "High"
end as classify
from sqlp_orders ;
```

Purpose:

This analysis is intended to divide orders into different value groups and count how many orders fall into each group by year. It helps understand the distribution of low, medium, and high-value orders.

Q13. Top-Performing Shop in Each State

```
select * from
(
  select s.shop_name,s.state, round(sum(o.total_amount),2) as total_amount,
  dense_rank() over(partition by s.state
  order by sum(o.total_amount) desc) as rnk
  from sqlp_shops as s
  join sqlp_orders as o
  on s.shop_id = o.shop_id
  group by s.shop_name, s.state) t
where rnk = 1;
```

Purpose:

This analysis identifies the shop with the highest revenue in each state. It helps compare shop performance within different geographical locations.

Q14. Customers Spending Above the Average

```
select customer_id, round(sum(total_amount),2) as total_amount from sqlp_orders
group by customer_id
having sum(total_amount) > (select round(avg(total_amount),2) as average from sqlp_orders)
and count(order_id) >= 1
order by customer_id asc;
```

Purpose:

This analysis identifies customers whose total spending is higher than the average amount calculated from the order data. It helps recognize higher-value customers.

Q15. Highest and Lowest Performing Shops by Shop Type

```

select * from
(
  select s.shop_type, s.shop_id, count(o.order_id) as order_delivered ,
  dense_rank() over(partition by s.shop_type
  order by count(o.order_id) asc) as lowest_rnk,
  dense_rank() over(partition by s.shop_type
  order by count(o.order_id) desc) as highest_rnk
  from sqlp_shops as s
  join sqlp_orders as o
  on s.shop_id = o.shop_id
  where o.order_status = "Delivered"
  group by s.shop_id,s.shop_type) t
  where highest_rnk = 1 or lowest_rnk = 1;

```

Purpose:

This analysis identifies the shops with the highest and lowest number of delivered orders within each shop type. It helps compare performance among similar types of shops.

Q16. Customers Placing Multiple Orders on the Same Date

```

select customer_id, count(order_id) , order_date from sqlp_orders
group by order_date, customer_id
having count(order_id) >= 2;

```

Purpose:

This analysis identifies customers who placed more than one order on the same date. It helps detect repeated or batch purchasing activity.

Q17. Three-Month Moving Average of Orders

```

with monthly_orders as (select shop_id,
month(order_date) as month_,
count(order_id) as total_orders
from sqlp_orders
group by shop_id, month(order_date))

select shop_id,month_,total_orders,round(avg(total_orders) over(partition by shop_id
order by month_ rows between 2 preceding and current row),2) as three_month_moving_avg

from monthly_orders
order by shop_id, month_;

```

Purpose:

This analysis is intended to calculate the average number of orders over a three-month period for each shop. It helps smooth short-term changes and observe overall ordering patterns.

Q18. Revenue Contribution by Product Category

```

with total_amt as ( select p.category , round(sum(o.total_amount)) as total_value
from sqlp_orders as o
join sqlp_products as p
on o.product_id = p.product_id
group by p.category)

select category,total_value,
round(total_value/(select sum(total_amount) from sqlp_orders)*100,2) as persenatge
from total_amt
where round(total_value/(select sum(total_amount) from sqlp_orders)*100,2) > 10;

```

Purpose:

This analysis calculates how much each product category contributes to the company's total revenue. It focuses on categories that contribute more than 10 percent of total revenue.

Q19. Customers Who Purchased from Every Category

```

select o.customer_id from sqlp_orders as o
join sqlp_products as p
on o.product_id = p.product_id
group by customer_id
having count(distinct p.category) =
(select count(distinct category) from sqlp_products);

```

Purpose:

This analysis identifies customers who have purchased products from every available category. It helps recognize customers with very broad purchasing behavior.

Q20. Busiest Day of the Week for Each Shop

```

select * from
(select s.shop_id,s.shop_name, dayname(o.order_date) as day_name,count(o.order_id),
rank() over(partition by s.shop_id order by count(o.order_id) desc) as rnk
from sqlp_orders as o
join sqlp_shops as s
on o.shop_id = s.shop_id
group by day_name, s.shop_id,s.shop_name) as t
where rnk = 1;

```

Purpose:

This analysis identifies the day of the week when each shop receives the highest number of orders. It can help understand peak business days for different shops.

Q21. Longest Gap between Customer Orders

```

select customer_id, max(gap) as max_gap from
(select customer_id ,datediff(order_date , lag(order_date)
over(partition by customer_id order by order_date)) as gap
from sqlp_orders) as t
group by customer_id
order by max_gap desc
limit 10;

```

Purpose:

This analysis measures the longest number of days between consecutive orders for customers. It helps identify customers who had long gaps between purchases.

Q22. Overall and State-Level Revenue Ranking of Shops

```
with s_table as (select s.shop_id,s.state ,round(sum(o.total_amount),2) as s_amount ,
dense_rank() over(partition by s.state order by sum(o.total_amount) desc) as s_rnk
from sqlp_orders as o
join sqlp_shops as s
on o.shop_id = s.shop_id
group by s.shop_id, s.state),

table_ as (select s.shop_id, round(sum(o.total_amount),2) as amount,
dense_rank() over(order by sum(o.total_amount) desc) as rnk
from sqlp_orders as o
join sqlp_shops as s
on o.shop_id = s.shop_id
group by shop_id)

select table_.shop_id as shop_id , table_.rnk as overall_rank,
s_table.s_rnk as state_rank ,table_.rnk - s_table.s_rnk as difference from table_
join s_table
on table_.shop_id = s_table.shop_id
order by table_.shop_id asc;
```

Purpose:

This analysis compares the revenue ranking of each shop at the overall level and within its respective state. It helps understand how a shop performs locally compared with its overall position.

Q23. Complete Customer Summary Report

```

with fav_cat as (select * from
(select o.customer_id , count(o.order_id) as order_count, p.category,
row_number() over(partition by o.customer_id order by count(o.order_id) desc) as rnk
from sqlp_orders as o
join sqlp_products as p
on o.product_id = p.product_id
group by o.customer_id,p.category) as t
where rnk = 1),

fav_pay as (select * from
(select customer_id , count(order_id) as order_count, payment_method,
row_number() over(partition by customer_id order by count(order_id) desc) as rnk
from sqlp_orders
group by customer_id,payment_method) as t
where rnk = 1)

select o.customer_id,concat(c.first_name," ",c.last_name) as full_name,
c.loyalty_tier,count(o.order_id) as total_order ,round(sum(o.total_amount)),
round(avg(o.total_amount)) as avg_value,f.category,p.payment_method
from sqlp_orders as o
join sqlp_customers as c
on o.customer_id = c.customer_id
join fav_cat as f
on f.customer_id = o.customer_id
join fav_pay as p
on p.customer_id = o.customer_id
group by o.customer_id ,
c.first_name,
c.last_name,
c.loyalty_tier,

```

Purpose:

This analysis creates a detailed summary of each customer, including their name, loyalty tier, number of orders, total spending, average order value, preferred category, and most-used payment method. It provides a complete overview of customer purchasing behavior.

Key Findings

The SQL analysis provides useful information about different aspects of the business. The analysis helps identify high-spending customers, best-selling products, inactive customers, top-performing shops, and important revenue patterns.

It also provides insights into customer purchasing behavior, loyalty tiers, product preferences, payment methods, and order frequency. Shop-level analysis helps compare performance across different shop types and states.

The analysis can support better business decisions by identifying strong-performing areas as well as products, customers, or shops that may require additional attention.

Conclusion

This Exploratory Data Analysis project shows that the business generates strong overall revenue (₹6.78 crore across 800 orders), with ₹4.21 crore (62%) coming from Delivered orders alone. However, nearly 17% of orders are Cancelled or Returned, with three shops — Nagpur Plaza (30.4%), Lucknow Central (29.1%), and Kochi Square (21.9%) — driving a disproportionate share of these problem orders. Beauty (22.3%), Electronics (21.9%), and Fashion (19.8%) are clearly the revenue engines of the business, together contributing close to two-thirds of total sales, while categories like Sports (3.4%) and Home & Kitchen (5%) remain underperforming and may benefit from targeted promotions. Best-selling products such as Nivea Face Wash and Nike Running Shoes (80 units each) confirm that Beauty and Fashion also lead in unit demand, not just revenue.

Customer analysis reveals a small set of high-value customers (led by Curtis Smith at ₹10.48 lakh and David Allen at ₹10.21 lakh) who anchor a large share of revenue, alongside a broader base of 100 out of 120 customers (83%) who shop across more than 3 different outlets rather than staying loyal to one. However, no customer purchases across all 8 product categories, and some customers have gone over 900 days without ordering again — pointing to real opportunities in cross-selling and win-back campaigns. Gold-tier customers show the highest average order value (₹95,532), narrowly ahead of Platinum (₹94,064), suggesting loyalty tier alone doesn't fully predict spending behavior. Online Hub emerges as the best-performing shop format, generating ₹2.82 crore in revenue from 333 orders — well ahead of Franchise Outlets and Flagship Stores — while Pop-up Stores lag far behind at just ₹38.6 lakh, suggesting the business's digital channel is outperforming physical formats and may deserve further investment.

Overall, the analysis demonstrates how SQL-driven exploration can convert raw transactional data into concrete, actionable business insights — identifying where revenue is concentrated, where operational issues (cancellations/returns) are hurting performance, which customer segments and shop formats drive the most value, and where growth opportunities (cross-category selling, shop-level fixes, seasonal promotions, and win-back campaigns) exist to strengthen the business going forward.